Multiple-Test Admissions

A joint analysis and corrected version of two Grady Towers letters

31 October 2026

Grady Towers's first letter gives incorrect overlap probabilities for correlated IQ tests; his follow-up explicitly retracts that calculation and introduces a substantially better numerical method. This report assesses the two together. At correlation 0.7, a bivariate-normal model with two top-2% cutoffs gives 3.3724% eligible through either test and 31.3814% repeat qualification among passers of one specified test. For top-2% and top-0.1% cutoffs on different tests, 65.8589% of higher-threshold passers also pass the lower-threshold test. The follow-up's remaining numerical differences arise chiefly from grid boundaries, rounded cutoffs, and inconsistent marginal denominators. Neither letter establishes actual membership proportions or a latent-ability interpretation. A corrected combined article and reproducible calculations are provided.

1 Joint assessment of the two sources

1.1 Read the second letter as a retraction

The sources are *On using multiple tests for high IQ society admissions* (Towers) and its *Followup* (Towers, 2000), both hosted by Darryl Miyaguchi. The first is undated in the supplied page. The second identifies its underlying letter as dated 12 January 2000 and says that it was published posthumously with permission. The supplied Wayback captures are dated 9 March 2026 and 5 May 2022, respectively; these are archive dates, not the letters' publication dates.

The follow-up expressly withdraws the earlier calculation. It should not be presented as independent support for the first letter's dramatic conclusion. Towers's useful qualitative point survives: accepting a pass on either of two imperfectly overlapping tests generally enlarges the eligible group relative to either test alone. The numerical examples and the interpretation of eligibility need correction.

This report examines the mathematical scenarios in those historical letters. It does not establish the admission policies, test correlations, or membership composition of any named society, either historically or today. The incomplete 1960 reference in the first letter can be identified as Zelen and Severo (1960); the calculations here were performed independently.

1.2 The first letter's principal numerical claims fail

Table 1 compares the first letter's principal figures with recalculations under its intended Gaussian scenario. In the mixed case, A means passing the top-2% cutoff on test X, and B means passing the top-0.1% cutoff on a different test Y. All corrected entries assume population score correlation $\rho = 0.7$.

Table 1: Selected claims in the original letter and corrected model results. All displayed probabilities are percentages, with the denominator specified.

Quantity	**Original**	**Corrected**
Both top-2% tests: population	about 0.2	0.627627
Either top-2% test: population	3.83	3.372373
Second top-2% pass, given first pass	10.27	31.381370
Mixed case: either pass, population	2.11	2.034141
Mixed case: both pass, population	about 0.014	0.065859
Mixed case: higher cutoff given lower, $P(B \mid A)$	0.69	3.292944
Mixed case: lower cutoff given higher, $P(A \mid B)$	13.48	65.858880

One error is visible without a distributional model: the mixed-case union cannot be 2.11% when its marginal probabilities sum to only 2.10%. In general, $P(A \cup B) \leq P(A) + P(B)$. The corrected mixed-case failure rate on the lower-threshold test among higher-threshold passers is 34.1411%, rather than the original's approximately 86%.

For the equal top-2% case, passing X alone has probability 1.3724%, and passing Y alone has the same probability. Passing *exactly one* therefore has probability 2.7447%. These are different quantities; wording such as "one test but not the other" should specify which is intended.

1.3 What the follow-up fixes, and what remains

The follow-up replaces an approximation involving the phi coefficient with numerical summation of a bivariate-normal density. That is a substantial correction. Reimplementing its supplied BASIC grid closely reproduces its published joint probabilities. The remaining issues are:

- The equal-weight half-IQ-point grid samples the cutoff itself, giving approximately the boundary of a midpoint cell a quarter point below that cutoff. It does not integrate exactly from the stated continuous cutoff.
- The integer IQ cutoffs 133, 150, and 164 do not have exactly the nominal tail probabilities used as denominators in the follow-up.
- A conditional proportion among passers of one specified test is not automatically a proportion among society members, or even among all people eligible through either of two tests.

Section A separates those effects. The absent scale and cell-area factors in the density ratio cancel; they are not fatal normalization errors. The image of the follow-up's double integral was also inspected, in addition to its HTML text and BASIC listing.

The follow-up recounts a top-5% example attributed to page 306 of Cattell's *The Scientific Analysis of Personality*. Under the model used here, two such tests at $\rho = 0.7$ have joint probability 1.95993%, rather than 1%. This checks Towers's recollection as written. Cattell's cited page was not independently inspected, so no error is attributed to Cattell himself.

2 Corrected combined article: admissions through correlated tests

An admission rule and a test percentile answer different questions. A test cutoff may be exceeded by 2% of a reference population. A rule accepting either of two such cutoffs concerns the union of two sets of people, whose size depends on their overlap. Requiring both cutoffs concerns their intersection. Neither rule, by itself, identifies the people with the highest values of a perfectly measured underlying ability.

2.1 Specify a joint model and distinguish the correlations

Let X, Y denote standardized scores in the same reference population, with a bivariate-normal distribution:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right], \qquad -1 < \rho < 1. \tag{1}$$

Here ρ is the Pearson correlation of the continuous scores. It is not the reliability of either test and is not generally the correlation of the binary pass/fail indicators. Normal marginal distributions and a correlation alone do not establish equation (1); the joint distribution is an additional assumption, especially consequential in the tails.

For desired marginal pass probabilities p_A, p_B, use

$$a = \Phi^{-1}(1 - p_A), \qquad b = \Phi^{-1}(1 - p_B), \qquad A = \{X > a\}, \quad B = \{Y > b\}. \tag{2}$$

The joint pass probability follows by conditioning on X:

$$\boxed{J = P(A \cap B) = \int_a^\infty \varphi(x)\, \overline{\Phi}\left(\frac{b - \rho x}{\sqrt{1 - \rho^2}}\right) dx.} \tag{3}$$

Here φ is the standard-normal density and $\overline{\Phi} = 1 - \Phi$ its upper-tail function. The remaining quantities are

$$P(A \cup B) = p_A + p_B - J, \tag{4}$$

$$P(B \mid A) = J/p_A, \qquad P(A \mid B) = J/p_B. \tag{5}$$

The denominators determine the question being answered.

For binary indicators $1_A, 1_B$, the phi coefficient is

$$\phi_{\text{binary}} = \frac{J - p_A p_B}{\sqrt{p_A(1 - p_A)p_B(1 - p_B)}}. \tag{6}$$

This differs from the correlation of the underlying continuous Gaussian variables (Ekström, 2011). For equal top-2% thresholds at $\rho = 0.7$, $\phi_{\text{binary}} \approx 0.29981$, rather than 0.7. The follow-up attributes its earlier error to a phi/product-moment approximation. Without the first algorithm, its exact computational mistake cannot be reconstructed; substituting one correlation for the other is nevertheless unjustified.

2.2 Two tests with the same nominal cutoff

For two top-2% cutoffs and $\rho = 0.7$, equation (3) gives

$$J = 0.0062762740205, \qquad P(A \cup B) = 0.0337237259795. \tag{7}$$

Thus 0.627627% of the population passes both tests, and 3.372373% passes at least one. Among people who pass a specified first test,

$$P(B \mid A) = \frac{0.0062762740205}{0.02} \approx 31.3814\%. \tag{8}$$

This statement concerns repeat qualification, not test reliability and not an unconditional probability of joining a society.

If eligibility instead means passing *either* of the two tests, the fraction of that eligible union passing both is

$$P(A \cap B \mid A \cup B) = \frac{J}{2p - J} \approx 18.6109\%, \qquad p = 0.02. \tag{9}$$

The 31.3814% and 18.6109% figures are both correct under the model; they use different denominators. Actual members may be a further selected subset of either group.

The same calculation at rarer equal cutoffs yields table 2. The cutoffs are exact nominal percentiles rather than rounded IQ points. IQ-style values in the table use mean 100 and standard deviation 16; they are scale conversions.

Table 2: Equal nominal cutoffs at $\rho = 0.7$: population overlap and repeat qualification among passers of one specified test.

Marginal tail on each test	**z cutoff**	**SD-16 cutoff**	**Both: population %**	**Second pass given first: %**
2%	2.053749	132.85998	0.627627	31.381370
0.1%	3.090232	149.44372	0.0159589	15.958919
1/33,000	4.010439	164.16702	0.000231250	7.631243

2.3 A lower cutoff on one test and a higher cutoff on another

Let $p_A = 0.02$ and $p_B = 0.001$, still with $\rho = 0.7$. The joint probability is

$$J = 0.0006585888039. \tag{10}$$

Consequently,

$$P(B \mid A) = 3.292944\%, \qquad P(A \mid B) = 65.858880\%, \tag{11}$$

$$P(A^c \mid B) = 34.141120\%, \qquad P(A \cup B) = 2.034141\%. \tag{12}$$

The asymmetry is expected: the higher-threshold passers are a much smaller group. These two events concern *different tests*. If both cutoffs were applied to the same score, the higher-cutoff event would be contained in the lower-cutoff event and the corresponding conditional probability would be 100%.

The result is sensitive to the assumed correlation. For these same mixed cutoffs, $P(A \mid B)$ is approximately 33.59% at $\rho = 0.5$, 65.86% at 0.7, and 97.99% at 0.9. A nominal 0.7 is an illustrative parameter, not a

universal empirical property of IQ tests. Population, norms, age, constructs, administration, and selection all matter.

These conditional model results cannot determine whether someone who passed a particular magazine test could join a particular society. That requires evidence about the actual instruments and the society's accepted routes. Nor does the model establish the earlier claim about the prior memberships of most high-scoring society members.

2.4 Eligibility size, rarity equivalents, and calibrated rules

The population fraction admitted by a rule can be expressed as a normal rarity equivalent. Under the equal top-2% example, the OR and AND fractions map to approximately 129.26 and 139.94, respectively, on an SD-16 scale:

$$I_{\text{rarity}} = 100 + 16\Phi^{-1}(1 - p_{\text{eligible}}). \tag{13}$$

This only matches a selected fraction. It does not show that the rules select the same people as a single test at that cutoff, establish "super-high reliability," or identify a particular level of latent intelligence. The first letter's corresponding AND rarity equivalent of about 146 follows from its erroneous overlap probability.

Multiple tests do not make a 2% overall eligibility rule impossible. Within the stated model, one can choose the common cutoff a_* to solve

$$2\overline{\Phi}(a_*) - J(a_*, a_*; 0.7) = 0.02. \tag{14}$$

The solution is approximately $a_* = 2.270055$, or 136.32088 on an SD-16 scale. Each test then has marginal pass rate approximately 1.16021%, while the two-test OR rule has overall rate 2%. This is an example of calibrating a *rule*; using more than one test is not inherently incompatible with a specified eligible fraction.

2.5 Interpret the population model and the testing process

A useful report separates three questions: who exceeds a cutoff on one instrument; who is eligible under a specified admission procedure; and what the evidence supports about an individual's abilities. Agreement on one does not settle the others. Score interpretations and reference populations need explicit support (American Educational Research Association et al., 2014, Standards 1.1–1.4 and 5.8–5.9).

An eligibility probability refers to the designated population before application or membership selection. Correlations measured among applicants or members need not equal the population correlation used in equation (1). Selection can change the relationship, and adjustments depend on the selection mechanism (Sackett and Yang, 2000).

A procedure accepting the best result from repeated attempts also differs from a fixed two-test procedure. For m pass events, independently of any Gaussian model,

$$\max_i P(A_i) \le P\left(\bigcup_{i=1}^{m} A_i\right) \le \min\left(1, \sum_{i=1}^{m} P(A_i)\right). \tag{15}$$

If attempts were independent with identical pass probability p, the union would be $1 - (1 - p)^m$. Actual retests usually require a model of dependence and administration history; practice and coaching can also change score distributions (Hausknecht et al., 2007). An analysis should calibrate the procedure actually used, including its attempts, stopping rule, and accepted instruments.

Towers was right to ask how multiple admission routes affect eligibility, and right to withdraw a faulty calculation. A defensible replacement states the rule, the reference population, the joint-distribution model, the thresholds, and each conditional denominator. Its conclusions remain conditional on those specifications.

A Numerical audit of the follow-up

The BASIC listing uses $\rho = 0.7$, means 100, standard deviations 16, and grid nodes from Q through 199 in steps of $h = 0.5$, with the full-field denominator starting at 1. In mathematical notation its probability estimate is

$$\hat{J}_Q = \frac{\sum_{x\in\{Q,Q+h,\ldots,199\}} \sum_{y\in\{Q,Q+h,\ldots,199\}} g(x,y)}{\sum_{x\in\{1,1+h,\ldots,199\}} \sum_{y\in\{1,1+h,\ldots,199\}} g(x,y)}, \tag{16}$$

where g is proportional to the Gaussian density on the IQ-point coordinates. The full grid has $397^2 = 157{,}609$ nodes. A constant factor $1/(16 \cdot 16)$ and a cell area h^2 would cancel between the two sums. The corresponding absent scale factor also cancels in the displayed double-integral ratio in the follow-up.

Table 3: Reproduction of the follow-up's grid estimates. Entries are probabilities, not percentages. The last column integrates from the actual continuous cutoff Q, on both tests.

Q	**Published grid ratio**	**Reproduced grid ratio**	**Continuous joint probability**
133	0.0064061692	0.0064061681	0.0061136512
150	0.00014748463	0.00014748460	0.00013830591
164	2.6393845×10^{-6}	2.6393840×10^{-6}	2.4390159×10^{-6}

The reproduced and published grid values agree to relative differences of about 2×10^{-7}, but not to every printed digit. The exact cause of those small differences is not established. They are far smaller than the difference between the grid estimate and the continuous probability at the intended cutoff.

Equal weighting at the boundary includes a full contribution from the node at Q. Viewed as midpoint quadrature, its first cell begins at $Q - h/2 = Q - 0.25$. The effective lower boundary is therefore shifted downward. The finite upper boundary and ordinary quadrature error add smaller discrepancies in these examples. If IQ scores were explicitly rounded to integers before classification, a separate rounding rule would be needed; a quarter-point grid shift cannot simply be assumed to be that rule.

For continuous scores, the actual marginal exceedance probabilities at the stated integer cutoffs are those in table 4. Dividing the continuous joint tail by its matching marginal gives a consistent conditional probability.

Table 4: Consistent continuous calculations at the follow-up's integer SD-16 cutoffs, with $\rho = 0.7$. All probability columns are percentages.

Cutoff	**Marginal pass rate**	**Both: population**	**Second pass given first**
133	1.95800788	0.61136512	31.223834
150	0.08890253	0.01383059	15.557028
164	0.00316712	0.00024390	7.701043

The follow-up divides its grid values by 0.02, 0.001, and 0.000030303. Those are approximations to desired nominal tails, not matching marginal tails of the specific grid or continuous integer cutoffs. Its approximately 32%, 15%, and 9% summaries consequently mix conventions. For exact nominal percentiles, use

table 2; for actual continuous integer cutoffs, use table 4. Using the grid's own marginal denominator gives approximately 31.51%, 15.74%, and 7.81%, a third, internally defined approximation.

B Reproducibility and numerical checks

The accompanying script evaluates equation (3) with adaptive one-dimensional quadrature and directly uses inverse survival functions for the thresholds. SciPy documents both its normal survival functions and its integration routine (SciPy contributors, n.d.a,n).

An independent integration identity supplies a cross-check:

$$J(a,b;\rho) = \overline{\Phi}(a)\overline{\Phi}(b) + \int_0^\rho \frac{\exp\left[-\dfrac{a^2 - 2tab + b^2}{2(1-t^2)}\right]}{2\pi\sqrt{1-t^2}}\,dt. \tag{17}$$

This follows by integrating the Gaussian probability's derivative with respect to its correlation, starting at independence. Checks also cover symmetry, probability bounds, the original mixed-case union violation, the binary phi coefficient, the calibrated OR threshold, and reproduction of the follow-up's grid.

The CSV files contain unrounded model calculations to complement the rounded display tables. Run `make check` from this analysis folder; `make pdf` builds the report with XeLaTeX. Numerical consistency does not establish the empirical truth of the Gaussian assumptions or the chosen correlation.

C Final verification results and limits

The final checks were executed on `computer10` on 31 October 2026. The Deal evidence report and the fresh SPARK campaign are pinned to the verified source revision. They retain exact commands, tool versions, statuses, source hashes, checksums, and raw logs. The Python environment used Python 3.13.5, Deal 4.24.6, and Deal Solver 0.1.2.

C.1 Deal checks across the repository

The Python results in table 5 cover both analyses and their shared verification tools; they are not separate totals for this Towers analysis. Every current Python file, including tests and fixtures, contains contracts, and every named function has a real Deal decorator.

Table 5: Final repository-wide Python verification results on computer10.

Check	Recorded outcome
Contract coverage	20 files, 142 named functions, 416 Deal decorators; zero gaps
Positive static lint	All 19 positive files: zero diagnostics, empty output, exit 0
Static negative control	Deliberately wrong fixture: exactly one expected `DEL012`, exit 1
Runtime and property tests	110 passed; zero failures or skips
Recorded commands	All 27 Python command checks reached their expected outcomes
Focused symbolic checks	Two pure proof-acceptance gate postconditions proved

Coverage, static lint, the numerical scripts, Ada-output validators, independent replays, and evidence audits also passed under `python -O`; CLI entry points explicitly enable Deal contracts. All 14 earlier production-linter diagnostics were resolved without suppressing findings or weakening the permitted exception families. Bounded quotient helpers retain the original division operations, and numerical tolerances are unchanged. The incorrect symbolic contract was rejected. Repository-root `make check` and GitHub Actions enforce coverage, clean positive lint, the expected negative diagnostic, and tests.

A clean linter result is not a universal proof of Python correctness. Runtime contracts and generated/property tests check executions and specified input domains. Deal Solver's successful symbolic result concerns only the two pure acceptance gates; it does not prove SciPy, the numerical validators, file handling, or the subprocess orchestrator. Skipped or unsupported checks are not counted as proofs.

C.2 SPARK core and numerical boundary

For this analysis, the fresh five-prover portfolio discharged **31 checks: 9 flow checks and 22 proof checks**, with zero unproved and zero justified checks. The latter comprise 13 run-time checks and 9 functional-contract checks. The native Towers report records all 11 subprograms/packages in `Admission_Core` analyzed. The deliberately incorrect union contract was rejected. CVC5, Z3, Alt-Ergo, classic COLIBRI, and Yices were requested together; completion of the portfolio does not mean that every independent solver run completed. Earlier incomplete independent runs remain recorded as incomplete. The recorded compiler reports GNAT 13.2.0 and the matching GNATprove binary reports banner `0.0w`; its identity is retained in the manifest.

Under the core's explicit input bounds and preconditions, these checks establish event classification, one-cell count updates, bounded marginal and union counts, and exact integer/rational probability algebra, including inclusion-exclusion, complements, and conditional numerators and denominators. They concern this specified core, rather than every mathematical or empirical claim in the report.

The floating-to-core adapter rounds probabilities to units of 10^{-15} and checks each reconstruction error against $\delta = 0.51 \times 10^{-15}$. For a conditional probability j/p, the resulting ratio change is bounded by $\delta(1 + j/p)/(p - \delta)$, when $p > \delta$. At the smallest nominal marginal $p = 1/33{,}000$, this is at most approximately 3.37×10^{-11}; actual checked changes are smaller. This accounts only for quantization. The driver and `Normal_Model` have `SPARK_Mode => Off`; `libm`, transcendental functions, inverse-normal calculation, Simpson quadrature, floating-point rendering, I/O, and random generation remain outside the proof. Comparing 10,000 and 20,000 Simpson panels is a numerical refinement check, not a proved quadrature error bound.

Independent validation compared the Ada calculations with SciPy and replayed the retained simulation from seed `20261030`. One million shared correlated pairs supplied the four Towers cases; every *neither, A only, B only, both* count matched exactly, both normally and under optimized Python. Replay checks implementation consistency; it does not prove random independence or the Gaussian model. Wilson intervals describe simulation uncertainty under that model, and rare event counts do not establish extreme-tail psychometric frequencies.

From a full Git clone, the evidence can be audited with `python verification/verify_evidence.py`; the pinned campaign report gives the command for auditing this fresh run. Audits check checksums, native totals, current Ada identities, and verified historical Python Git blobs. They do not independently recheck the SMT proofs or replace rerunning the provers. None of these software checks establishes that $\rho = 0.7$ applies to actual tests, that their norms and tails are accurate, or that eligibility probabilities describe real membership selection or latent ability. Those remain empirical and modeling questions identified in the main analysis.

References

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. *Standards for Educational and Psychological Testing*. American Educational Research Association, Washington, DC, 2014. ISBN 978-0-935302-35-6. URL https://www.testingstandards.net/uploads/7/6/6/4/76643089/9780935302356.pdf.

Joakim Ekström. The phi-coefficient, the tetrachoric correlation coefficient, and the Pearson–Yule debate. Technical report, UCLA Department of Statistics, 2011. URL https://escholarship.org/uc/item/7qp4604r.

John P. Hausknecht, Jane A. Halpert, Nicole T. Di Paolo, and Meghan O. Moriarty Gerrard. Retesting in selection: A meta-analysis of coaching and practice effects for tests of cognitive ability. *Journal of Applied Psychology*, 92(2):373–385, 2007. doi: 10.1037/0021-9010.92.2.373. URL https://ecommons.cornell.edu/entities/publication/97becf46-3c0a-4de4-a48f-79b715ba8446.

Paul R. Sackett and Hyuck Yang. Correction for range restriction: An expanded typology. *Journal of Applied Psychology*, 85(1):112–118, 2000. doi: 10.1037/0021-9010.85.1.112. URL https://pubmed.ncbi.nlm.nih.gov/10740961/.

SciPy contributors. *scipy.stats.norm*. SciPy, n.d.a. URL https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html. Reference documentation. Accessed 30 October 2026.

SciPy contributors. *scipy.integrate.quad*. SciPy, n.d.b. URL https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.quad.html. Reference documentation. Accessed 30 October 2026.

Grady Towers. On using multiple tests for high IQ society admissions. Letter hosted by Darryl Miyaguchi. URL https://web.archive.org/web/20260309223409/http://miyaguchi.4sigma.org/gradytowers/societies.html. Undated in the supplied page. Wayback capture: 9 March 2026, 22:34:09 UTC. Accessed 30 October 2026.

Grady Towers. Followup to "on using multiple tests for high IQ society admissions". Letter hosted by Darryl Miyaguchi; published posthumously with permission, 2000. URL https://web.archive.org/web/20220505012535/http://miyaguchi.4sigma.org/gradytowers/followup.html. Letter dated 12 January 2000. Wayback capture: 5 May 2022, 01:25:35 UTC. Accessed 30 October 2026.

Marvin Zelen and Norman C. Severo. Graphs for bivariate normal probabilities. *Annals of Mathematical Statistics*, 31(3):619–624, 1960. doi: 10.1214/aoms/1177705789. URL https://doi.org/10.1214/aoms/1177705789. Identifies the original letter's incomplete reference; metadata verified, full article not used for the calculations.